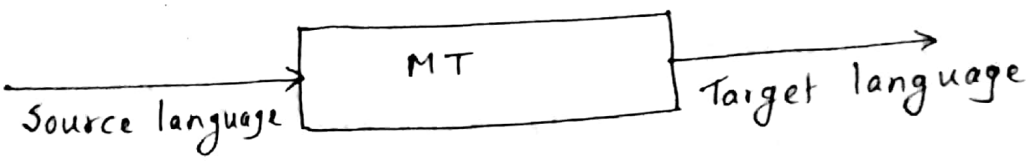
Dr. Mohammad Nurul Huda
statistical Machine Trans
Lecture sheet - I

Machine Translation (MT):

MT can be defined as the use of computers to automate some or all of the process of translating from one language to another.



MT uses ideas and techniques

- Linguistics
- Computer Science
- Artificial Intelligence (AI)
- Translation theory
- statistics

Commercially Interesting:

- US has invested in MT for intelligence purposes.
- EU spends more than €1,000,000,000 on translation costs each year.
- MT is used in UNL (Universal Network Language)
- MT is used to translate technical documents, reports, instruction manuals etc.

Objective:

- The goal is not fluent translation
- Only correct and understandable output.

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Various approaches to Machine Translation:

- The transfer approach
- The Interlingua approach
- The direct approach
- Corpus-based approach
- Word-for-word translation
- statistical translation

Word-for-word translation:

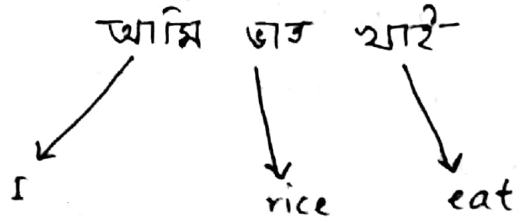
- Use a machine-readable bilingual dictionary to translate each word in a text.

Dictionary:

Bangla	English
আমি	I
ভাত	rice
খাই-	eat
.	.
.	.
.	.

Bangla: আমি ভাত খাই-

English: ?



English: I rice eat

Advantage:

Easy to implement, results give a rough idea about what the text is about.

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vantage:

Problems with word order means that this results in low-quality translation.

The Transfer Approach:

The transfer model involves three stages

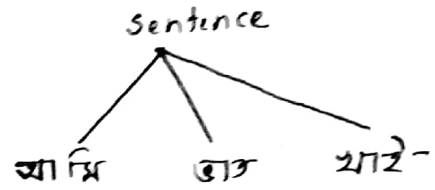
- Analysis
- transfer
- generation

Analysis:

The source language sentence is parsed, and the sentence structure and the constituents of the sentence are identified.

Example:

আমি ভাত খাই-

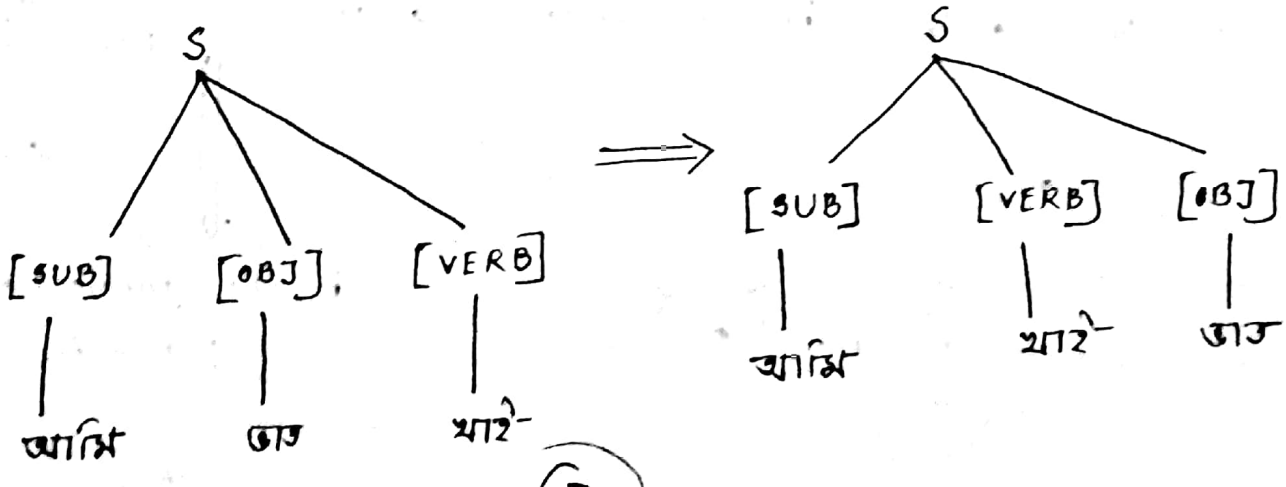


Words: আমি, ভাত, খাই-

Sentence structure: [subject] [object] [verb]

Transfer stage:

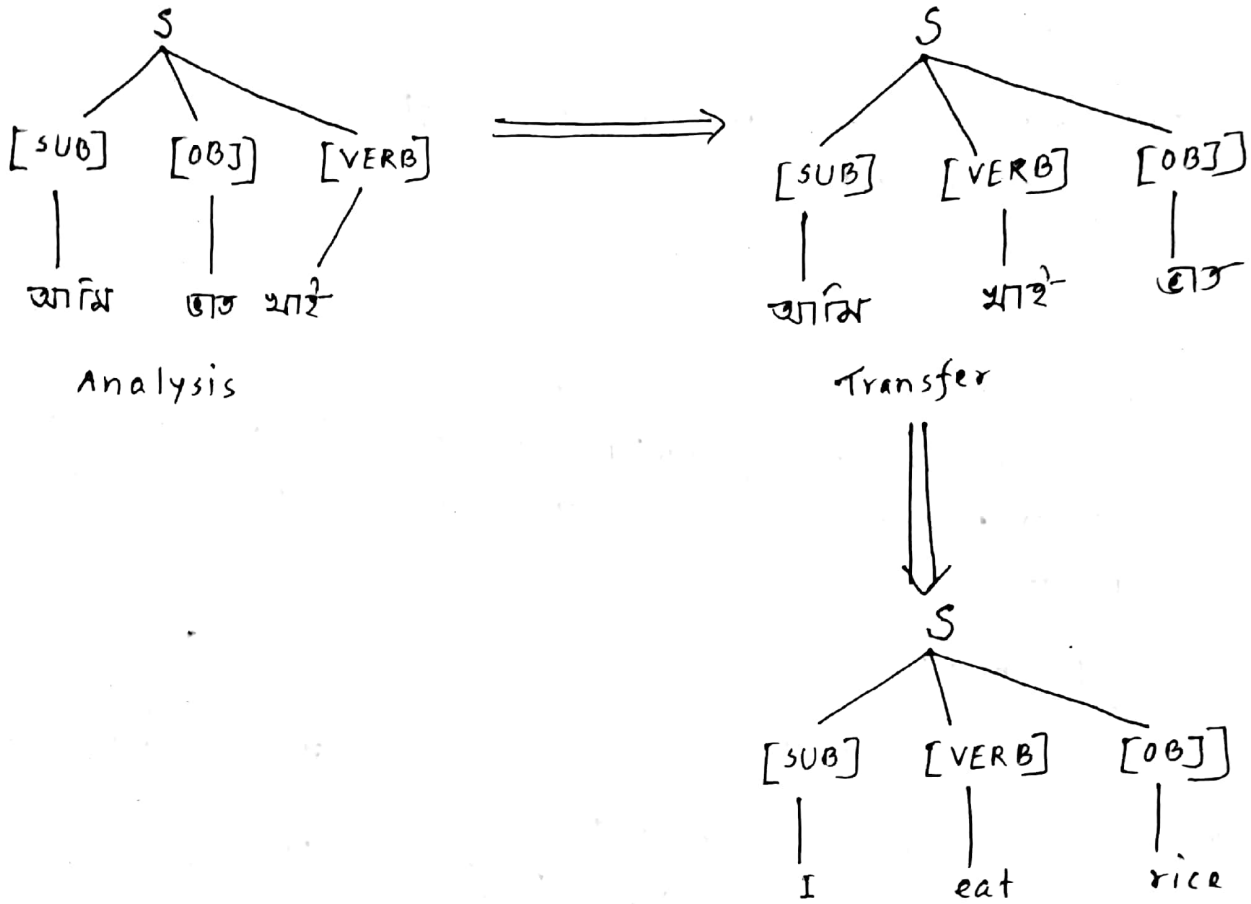
Transformations are applied to the source language parse tree to convert the structure to that of the target language.



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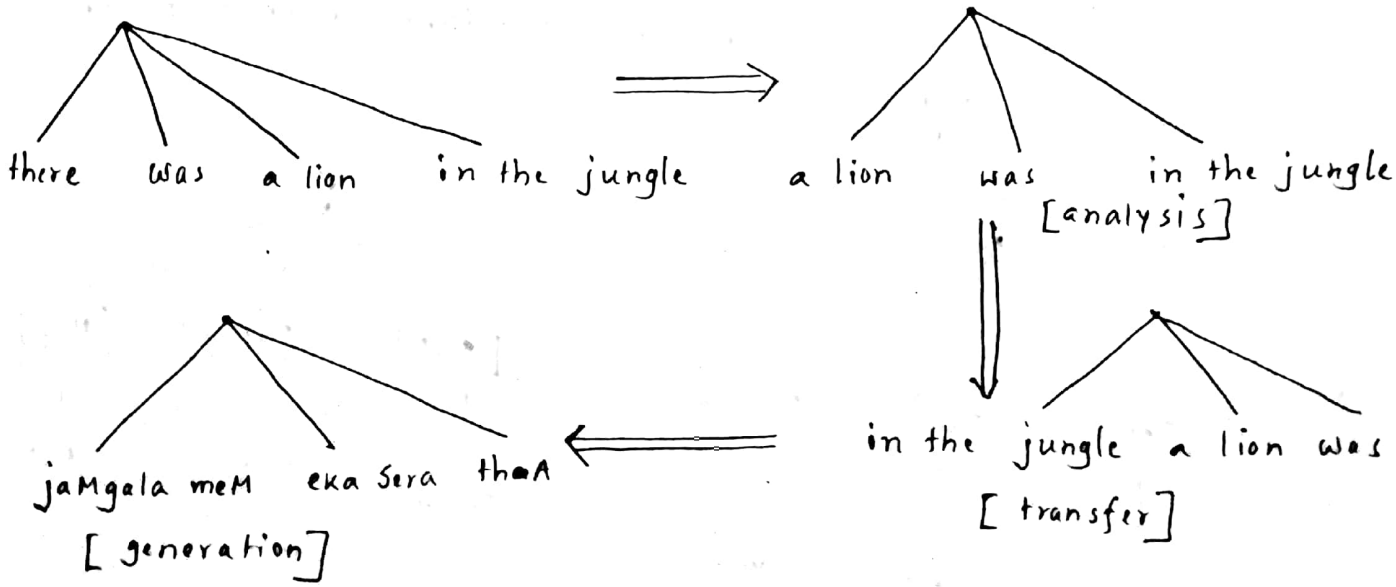
The generation stage :

Translates the words and expresses the tense, number, gena etc. in the taget language.



Example :

S : There was a lion in the jungle



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Interlingua Approach:

It is a two stage process.

1. extracting the meaning of a source language sentence in a language-independent form.
2. Generating a target language sentence from the meaning.

Importance:

It expresses the complete meaning of any sentence by using a set of universal concepts and relations.

Example:

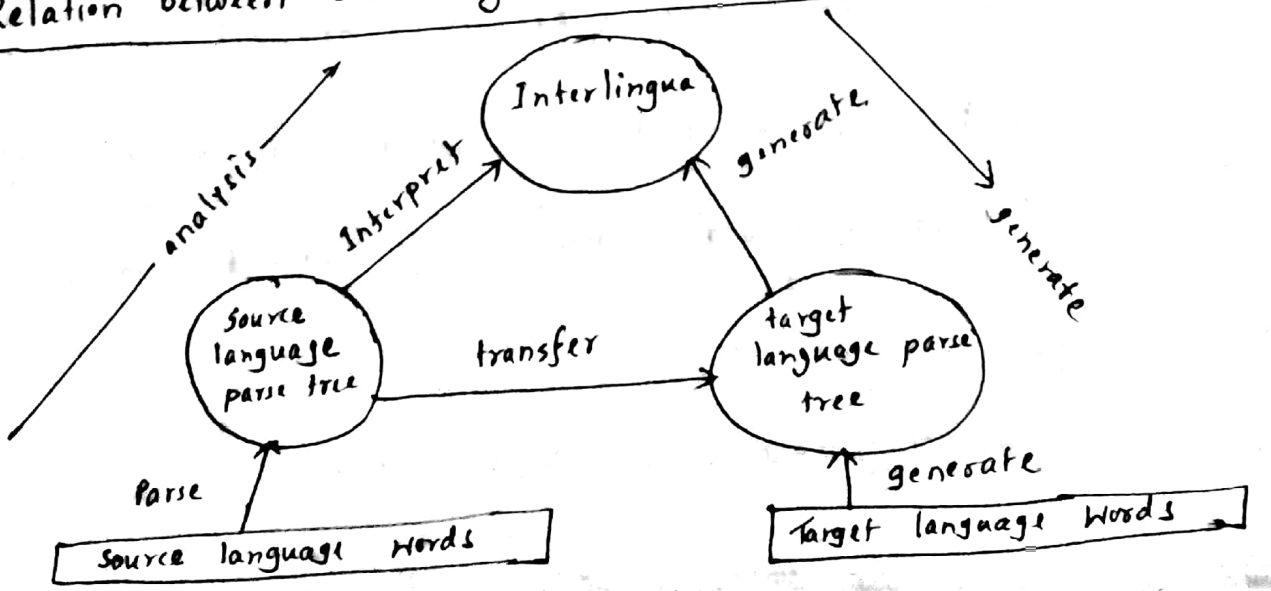
The child broke the toys with a hammer.

Interlingual Representation

PREDICATE	BREAK
AGENT	CHILD number: SING
THEME	TOY number: PLUR
INSTRUMENT	HAMMER number: SING
TENSE	PAST

⟨AGENT⟩ ⟨⟨PREDICATE⟩ ⟨TENSE⟩⟩ ⟨THEME⟩ ⟨INSTRUMENT⟩

Relation between Interlingua and Transfer approach:



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The Direct Approach:

step1: Morphological analysis

step2: Identify constituents

step3: Reorder them according to the constituent order in target language

step3: look up the words in an source-target language dictionary

step4: Inflect the target language words appropriately.

Example:

We will demand peace in the country.

English Sentence:

We will demand peace in the country.

Morphological Analysis:

We demand FUTURE peace in country

Identify constituents:

<We> <demand FUTURE> <peace> <in country>

Reorder according to target language:

<We> <in country> <peace> <demand FUTURE>

Look-up in the source-target language dictionary

<আমরা> <দেশের মধ্যে> <শান্তি> <দাবি FUTURE>

Inflect:

আমরা দেশের শান্তি দাবি করব।

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pus-Based Approaches:

- Not use human-encoded linguistic knowledge to solve the translation problem
- Used a training corpus of already translated texts - a parallel corpus - to guide the translation process.

A Parallel Corpus

Source language collection	Target language collection

Example Based Machine Translation:

- Split the problem into subproblems.
- Recall how they solved similar subproblems in the past.
- Adapt these solutions to the new situation
- Combine the solutions to solve the bigger problem.

Example:

He bought a huge house.

Corpus:

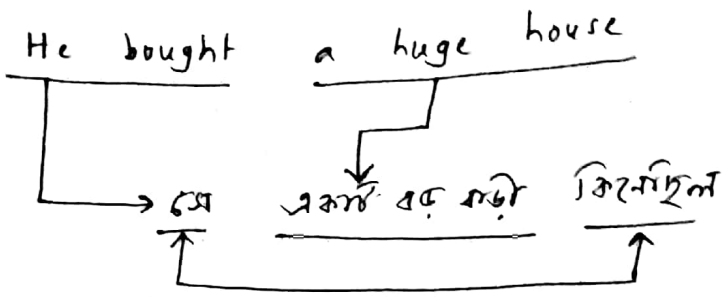
1a) He bought a pen

1b) সে একটি কলম কিনেছিল

2a) All ministers have a huge house

2b) সব মন্ত্রীর একটি বড় বাড়ি আছে

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Example based MT (EBMT) entails three steps:

1. Matching fragments against the parallel corpus.
2. Adapting the matched fragments to the target languege.
3. Recombing these translated fragments appropriately.

Why machine Translation is hard?:

- Word order
- Word sense
- Pronouns
- Tense
- Idioms

Word order:

English: SUB- VERB- OBJECT

Bangla: SUB- OBJECT- VERB

English: I eat rice

Bangla:

Sense ambiguity:

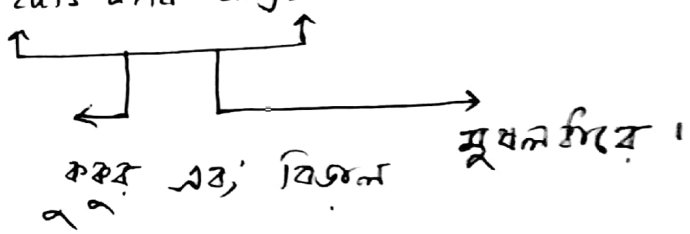
Example 1:
- Bank as in river
- Bank as in financial Institution

Example 2:
- Plant as in a tree
- Plant as in a factory

** Different word senses will likely translate into different words in another language.

Idioms:

It has been raining cats and dogs



Which one will be appropriate
- বিড়াল এবং কুকুর
- মুষলধারে।

Pronouns:

When should we use 'she' or 'he'.

সে ভাত খায় — He eats rice
সে ভাত খায় — she eats rice
} Which one is correct?

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